

Multimodality Imaging of Cholecystectomy Complications

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Abbreviations: ERCP = endoscopic retrograde cholangiopancreatography, FDG = fluorodeoxyglucose, MRCP = MR cholangiopancreatography

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Discuss common imaging findings of bile leaks.
- Identify the most common locations of retained stones and dropped stones.
- Describe imaging findings concerning for gallbladder cancer.

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Cholecystectomy is one of the most common surgeries performed in the United States. Although complications are uncommon, the high incidence of this surgery means that a radiologist will likely encounter these complications in practice. Complications may arise in the immediate postoperative period or can be delayed for weeks, months, or years after surgery. Vague and nonspecific symptoms make clinical diagnosis challenging. As a result, multimodality imaging is important in postoperative evaluation. US and multidetector CT are the usual first-line imaging modalities. Hepatobiliary scintigraphy, SPECT/CT, and MRI with conventional or gadopentate hepatobiliary contrast material are important and complementary modalities that are used for workup. The authors begin with a brief discussion of surgical technique and expected postoperative findings and then describe complications organized into four groups: (a) biliary complications, (b) stone-related complications, (c) iatrogenic complications, and (d) gallbladder complications. Biliary complications include bile leaks and bilomas, acute biliary obstruction, and biliary stricture. Stone-related complications include retained and recurrent stones and spillage of stones into the peritoneum. Iatrogenic complications include hemorrhage, vasculobiliary injury, arterial pseudoaneurysms, duodenal injury, and migration of clips. Gallbladder complications include recurrent cholecystitis after subtotal reconstituting cholecystectomy and unexpected gallbladder cancer.

An invited commentary by Mullens and Ibrahim is available online. Online supplemental material is available for this article.

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Introduction

Cholecystectomy is indicated for symptomatic diseases of the gallbladder and biliary tree, including symptomatic cholelithiasis, acute cholecystitis, choledocholithiasis, and biliary dyskinesia. Relative contraindications include untreated coagulopathy or advanced cirrhosis (1,2). Laparoscopic cholecystectomy is considered the standard of care for the treatment of gallstones. After this technique was developed by European surgeons in the 1980s, it was introduced to the broader general surgical community in the early 1990s and was quickly incorporated into general surgical practice within 2–3 years. The rapid adoption of this technique as a routine surgical procedure was unparalleled in surgical history and heralded the era of minimally invasive surgery. The success of this technique is owed to its demonstrated reduction in postoperative pain and length of hospital stay, rapid recovery, minimal scarring, and overall decreased medical costs (3,4). More than 750 000 cholecystectomies are now performed annually in the United States, the majority of which are

TEACHING POINTS

- In the first few weeks after surgery, postsurgical edema in the gallbladder fossa is common and should not be confused with complications that require treatment.
- Conventional biliary anatomy is seen in only 60% of the population; however, many biliary variants are unimportant to the performance of cholecystectomy. One of the most common surgically important variants is the right posterior hepatic duct draining into the common hepatic duct or into the cystic duct.
- CT and US of a bile leak often show a small to moderate amount of perihepatic fluid, generally accumulating around the right perihepatic space and tracking inferiorly into the right paracolic gutter. Occasionally, the fluid leaks predominantly around the left lobe of the liver and stomach.
- Heavily T2-weighted MRCP sequences must be performed before administering gadoxetate because it is excreted into the bile ducts as early as 5–10 minutes after administration and results in T2 shortening and loss of the signal intensity of fluid in the bile ducts, reducing the diagnostic quality of the MRCP images.
- Even in the absence of an identified stone nidus at imaging, an unexplained abscess or inflammatory mass in the hepatorenal fossa in a patient with a history of cholecystectomy should be considered highly suspicious for a dropped gallstone as the underlying cause.

performed with a laparoscopic approach, making it the most common gastrointestinal surgery (5). Most cholecystectomies are uneventful. However, laparoscopic cholecystectomy is known to have a higher rate of serious complications than the open techniques, particularly bile duct injury (6).

In 10%–40% of patients, biliary colic or right upper quadrant pain and a variety of other gastrointestinal symptoms persist after cholecystectomy; this phenomenon of continuation of symptoms (or development of new symptoms) is sometimes referred to as postcholecystectomy syndrome. There are multiple causes for this syndrome, including extrabiliary processes (eg, esophagitis, peptic ulcer disease, pancreatitis, and irritable bowel syndrome) or biliary processes (eg, biliary strictures, bile leaks, retained calculi, and other postsurgical complications), many of which are discussed in this article (7). Because of the wide range of possible causes of persistent or new symptoms after this definitive treatment of gallstones, there is increasing reliance on multimodality imaging in the evaluation of patients with nonresolved symptoms after cholecystectomy.

The overall complication rate of laparoscopic cholecystectomy is 1.6%–5.3% (8). The most common complications (eg, wound infection, urinary retention, and postoperative bleeding) are not unique to cholecystectomy and typically do not require imaging. The most common surgical complications requiring imaging are bile duct injury and retained common bile duct stones, occurring in

0.32%–0.52% and 0.5% of cases, respectively (8). Although complications of cholecystectomy are rare, the high frequency of this procedure means that a radiologist is likely to encounter them in practice. Complications at multimodality imaging may manifest in the immediate postoperative period or be delayed for weeks, months, or even years. Prompt recognition is important, because when they are unrecognized, these complications can result in severe morbidity or mortality (8). The complications can be conceptually organized into four groups: (a) biliary complications (eg, bile leakage and bilomas, acute biliary obstruction, and biliary stricture), (b) stone-related complications (eg, retained and recurrent stones and spillage of stones into the peritoneum), (c) iatrogenic complications (eg, hemorrhage, arterial pseudoaneurysm, duodenal injury, and migration of clips), and (d) gallbladder complications (eg, recurrent cholecystitis after subtotal reconstituting cholecystectomy and unexpected gallbladder cancer).

Laparoscopic Cholecystectomy Overview and Technique

After insufflating the abdomen and inserting the laparoscopic ports, the surgeon identifies the gallbladder and dissects it at the hepatocystic triangle (ie, the triangular space bordered by the cystic duct laterally, the common hepatic duct medially, and the inferior edge of the liver superiorly). The surgeon dissects the surrounding tissue and mobilizes the gallbladder from the cystic plate. The goal is to identify the critical view of safety, where the surgeon visualizes only two structures entering the gallbladder: the cystic duct and the cystic artery. If the critical view is identified, the surgeon clips the cystic duct and artery and removes the gallbladder. If the critical view of safety is not identified, the surgeon may convert the laparoscopic procedure to open surgery, attempt to clarify the anatomy with intraoperative cholangiography, or perform a subtotal cholecystectomy (9).

Intraoperative cholangiography is performed by inserting a cannula in the cystic duct and injecting contrast material under fluoroscopic guidance. This procedure can help to clarify the biliary anatomy and identify cystic duct or common bile duct stones and biliary injuries intraoperatively (2). However, routine use is controversial, and application of this procedure varies by institution.

A subtotal cholecystectomy is a well-described alternative (ie, “bail-out”) procedure used when the critical view of safety is not identified (10). Other indications for subtotal cholecystectomy are also related to factors that prevent safe removal of the total gallbladder, including severe cholecystitis (eg, gangrenous or perforated cholecystitis) and cirrhosis and portal hypertension (11). The

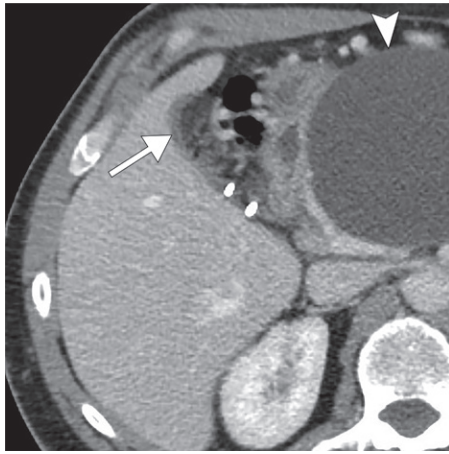


Figure 1. Expected findings in a 39-year-old man 8 days after laparoscopic cholecystectomy for recent gallstone pancreatitis. Axial contrast-enhanced CT image shows minimal fluid and fat stranding in the gallbladder fossa (arrow). A pancreatic pseudocyst is also partially seen (arrowhead).

two types of subtotal cholecystectomy are *subtotal fenestrating cholecystectomy* and *subtotal reconstituting cholecystectomy*. These terms have replaced the more ambiguous term *partial cholecystectomy*. For a subtotal fenestrating cholecystectomy, the peritonealized portion of the gallbladder is removed while the portion of the gallbladder adhered to the liver is left in situ. The remaining gallbladder mucosa is ablated, and the cystic duct is closed or ligated. This leaves an open nonfunctional gallbladder remnant and a closed cystic duct. For a subtotal reconstituting cholecystectomy, a portion of the gallbladder is removed, and the open portion of the gallbladder is closed to form a new lumen, usually after mucosal ablation. These two types of surgery are usually performed as open procedures but can be done laparoscopically, and there have been no studies directly comparing the two techniques (10).

Expected Immediate Postoperative Findings

In the first few weeks after surgery, postsurgical edema in the gallbladder fossa is common and should not be confused with complications that require treatment. At US, the fat in the gallbladder fossa may be hyperechoic and contain a small amount of nonorganized fluid. At CT and MRI, stranding and minimal fluid are generally confined to the gallbladder fossa (Fig 1). Uncommonly, hemostatic agents are applied to the surgical bed, which appear as bubbles of gas and mimic a fluid collection or a retained surgical sponge (Fig 2). When larger amounts of fluid are present in the gallbladder fossa, distinguishing normal postoperative fluid from a bile leak

or postoperative infection is difficult and may warrant additional imaging. Other commonly encountered expected postoperative findings include a small pneumoperitoneum and subcutaneous emphysema of the anterior abdominal wall, stranding, and ill-defined areas of fluid from placement of a laparoscopic port or subcostal incision (in open cholecystectomy) (12,13).

Biliary Complications

Biliary injuries are the most common complications after cholecystectomy. Bile leaks, bilomas, or acute biliary obstruction usually manifest soon after surgery, while biliary strictures often manifest months or years after surgery.

Biliary injuries are more common with laparoscopic cholecystectomy than with open cholecystectomy, occurring in 0.3%–0.5% of laparoscopic surgical procedures, and therefore, rates of injury have increased over time because of increasing adoption of a laparoscopic approach (8). The signs and symptoms of biliary injuries are often nonspecific, such as pain and fever with or without jaundice and pruritis. Elevated serum bilirubin or alkaline phosphatase levels are neither sensitive nor specific for a biliary injury (9). Thus, recognizing that many bile injuries may manifest without biliary ductal dilatation is important (14,15). Risk factors include severe gallbladder inflammation or gallbladder fossa adhesions, chronic cholecystitis, and variant biliary anatomy, which may obscure or alter the critical view of safety.

Conventional biliary anatomy is seen in only 60% of the population; however, many biliary variants are unimportant to the performance of cholecystectomy (16). One of the most common surgically important variants is the right posterior hepatic duct draining into the common hepatic duct or into the cystic duct (Fig 3). In one study of 73 cadavers, these variants were found to occur in 2.7% and 1.4% of individuals, respectively (17). Another important variant, a subvesical bile duct (previously termed the *duct of Luschka*) is prone to injury because of its close proximity to the surgical field (Fig 4). The precise definition of the subvesical duct is debated, but it can be considered a bile duct that runs in close proximity to the gallbladder fossa, either a superficial intrahepatic duct draining to the right hepatic duct, a bile duct draining directly into the gallbladder, or multiple small ducts in the connective tissue between the liver and gallbladder. The prevalence of a subvesical duct is reported to be 4%–10% of the population; however, these are rarely seen at imaging, presumably because of their small size (18). Congenital anatomic variants of the cystic duct occur in 18%–23% of the population and increase the risk of bile duct injury during

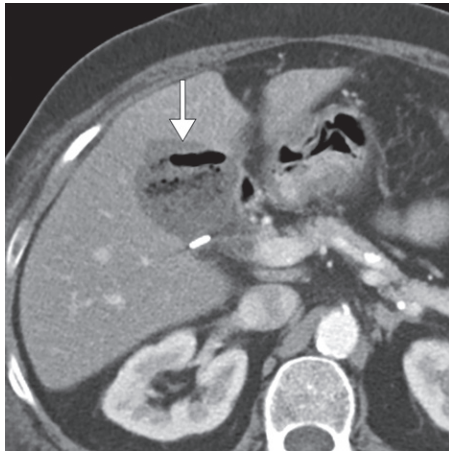


Figure 2. Hemostatic agent mimicking an abscess on day 3 after laparoscopic cholecystectomy in a 56-year-old woman. Axial contrast-enhanced CT image shows a heterogeneous collection of fluid and gas in the gallbladder fossa (arrow). The initial diagnosis was an abscess; however, the surgical report indicated that a hemostatic agent, which appears as multiple small gas bubbles, was applied.

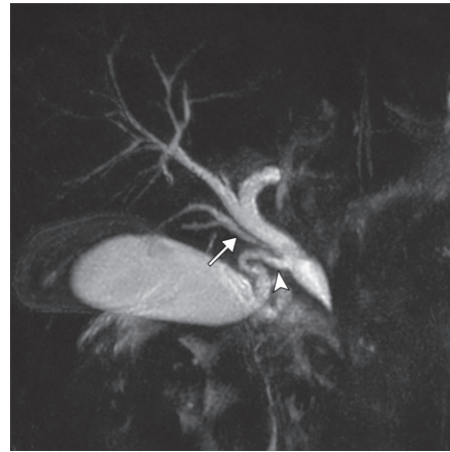


Figure 3. Variant anatomy with an aberrant posterior right hepatic duct. Coronal-oblique three-dimensional maximum intensity projection MR cholangiogram in a 57-year-old man with stricture of the common bile duct due to chronic pancreatitis shows the right posterior bile duct (arrow) with abnormally low insertion on the common duct, just superior to the cystic duct insertion (arrowhead). This makes the duct prone to misidentification and injury.

cholecystectomy. Common variants that increase the risk for injury are a low insertion of the cystic duct, a long parallel course of the cystic duct to the common hepatic duct, and a short cystic duct (19,20).

These variant ducts may lead to misidentification of the cystic duct, inadvertent bile duct transection, or thermal injury, which may not be recognized intraoperatively (16,21). In the surgical literature, there are multiple classification systems for biliary injuries. The older ones, such as the Bismuth and Strasberg classification systems, were based on the location of the injury, and the newer ones, such as the Stewart-Way and Hannover classification systems, incorporate the mechanism of injury (13,15,22). Regardless of the classification system used at individual institutions, accurate anatomic description and localization of the injury are most important to guide treatment.

Bile Leaks and Bilomas

Bile leaks occur in the immediate postoperative period and most often arise from the cystic duct and the subvesical bile duct (9). The leak may occur because of a slipped cystic duct clip or a dissection injury secondary to variant biliary anatomy (15,16,18). Most patients present with abdominal pain, and only a minority present with biliary peritonitis (15,23). If a bile leak is contained as a discreet collection of fluid, it is termed a *biloma*. Bilomas are usually located near the site of injury, classically in the gallbladder fossa, perihepatic space, or subcapsular space of the liver. These fluid collections have a

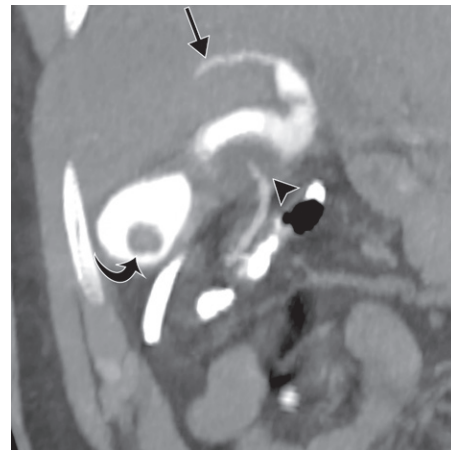


Figure 4. Variant anatomy with a subvesical duct (duct of Luschka) in a 32-year-old man. The patient underwent attempted cholecystectomy and the surgeon discovered an unusual biliary ductal anatomy, so a cholecystostomy tube was placed and cholangiography and CT were performed. Coronal noncontrast maximum intensity projection CT image shows the anterior right biliary duct (straight arrow) draining into the neck of the gallbladder, which is consistent with a subvesical duct. A stone is seen in the gallbladder (curved arrow) as well as the balloon of the cholecystostomy tube (arrowhead).

high risk of infection after cholecystectomy for acute cholecystitis (24).

CT and US of a bile leak often show a small to moderate amount of perihepatic fluid, generally accumulating around the right perihepatic space and tracking inferiorly into the right paracolic gutter (Fig 5). Occasionally, the fluid leaks predominately

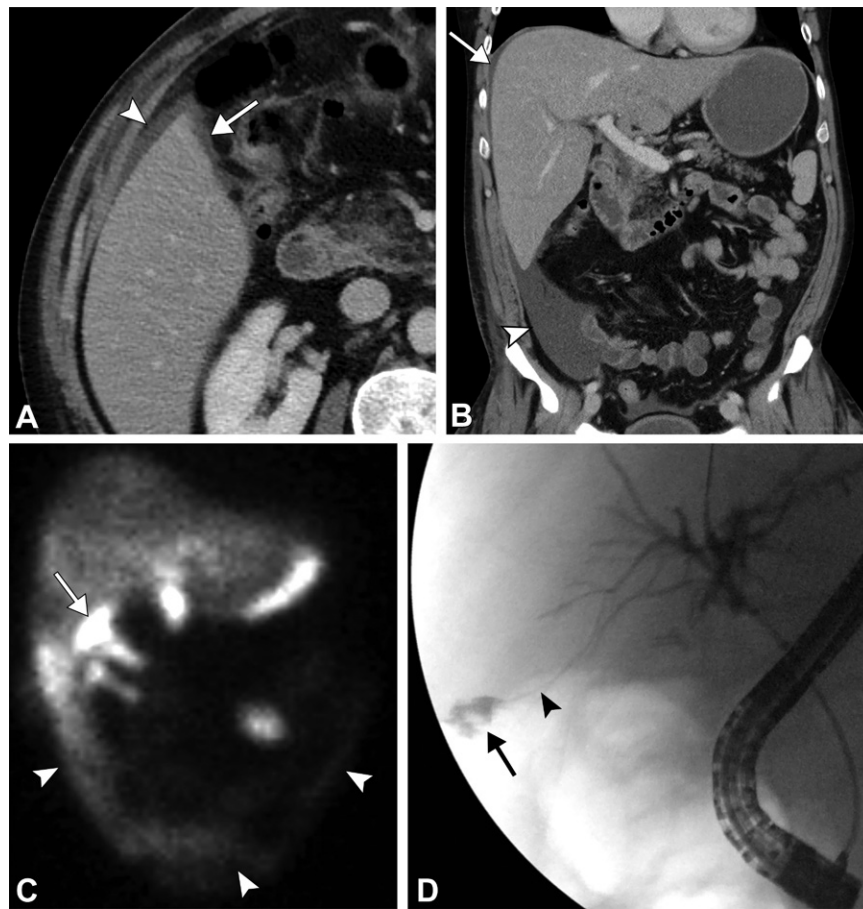


Figure 5. Bile leak in a 46-year-old man who presented with generalized abdominal pain 5 days after cholecystectomy. (A) Axial contrast-enhanced CT image shows trace free fluid in the gallbladder fossa (arrow), which is contiguous with trace perihepatic free fluid (arrowhead). (B) Coronal contrast-enhanced CT image shows the perihepatic free fluid (arrow) connecting with a larger area of free fluid that is predominately accumulating in the right paracolic gutter (arrowhead). (C) Left anterior-oblique technetium ^{99m}Tc hepatobiliary iminodiacetic acid (HIDA) scintigraphic image at 45 minutes shows accumulation of tracer in the gallbladder fossa (arrow) and a diffuse bile leak in the paracolic gutters and pelvis (arrowheads). (D) Posteroanterior fluoroscopic image obtained during endoscopic retrograde cholangiopancreatography (ERCP) before stent placement shows contrast material leaking into the peritoneal space at the gallbladder fossa (arrow), emanating from the subvesical bile duct (arrowhead).

around the left lobe of the liver and stomach. Therefore, the radiologist may be the first practitioner to suggest a bile leak when asymmetric abdominal free fluid is encountered after cholecystectomy. CT and US of a biloma demonstrate a simple or loculated fluid collection with an enhancing wall of variable thickness. The imaging appearance of a biloma lacks specificity because the findings overlap with other postoperative fluid collections such as abscesses and seromas. Technetium ^{99m}Tc hepatobiliary iminodiacetic acid (HIDA) scintigraphy has high specificity for bile leaks and bilomas but lacks the spatial resolution to allow their locations to be determined (Fig 6). SPECT/CT can improve localization of ^{99m}Tc HIDA in a fluid collection when compared with that with planar scintigraphy alone (21). In a study (25) of 32 patients with a suspected bile leak, planar HIDA scintigraphy had an accuracy of 65.6%, while the addition of SPECT/CT increased the accuracy to 96.8%.

The combination of MR cholangiopancreatography (MRCP) and gadoxetate disodium (Eovist; Bayer)–enhanced MRI has advantages over CT or HIDA scintigraphy because it is highly sensitive and specific for identifying leaks and allows accurate identification of the biliary anatomy and the location of the leak (Fig 7) (21,26). MRCP

without contrast material allows accurate anatomic evaluation of the fluid-filled bile ducts and provides indirect evidence of a leak. The addition of gadoxetate-enhanced sequences can provide functional information and direct evidence of a leak in the form of contrast-enhanced bile.

In a healthy person, 50% of the administered gadoxetate dose is excreted in the biliary system, which results in T1-weighted shortening of fluid in the biliary system, which is best identified during the hepatobiliary phase of contrast-enhanced MRI. The hepatobiliary phase occurs 10–120 minutes after contrast material injection. Most of the confirmatory data have involved imaging during the hepatobiliary phase after a 20-minute delay, which is usually sufficient for visualization of the bile duct (27,28). However, a standard 20-minute delay may be inadequate for a variety of reasons. First, biliary excretion of gadoxetate is related to hepatocyte function, so further delayed images should be considered in patients with abnormal liver function test results. Second, a bile leak may not be evident with a 20-minute delay, even if the intrahepatic bile ducts are opacified (14,29).

In one study (29) of 34 patients, the sensitivity for identification of a bile leak was 42.9% for a 20–25-minute delay and 96.4% for a 60–90-minute

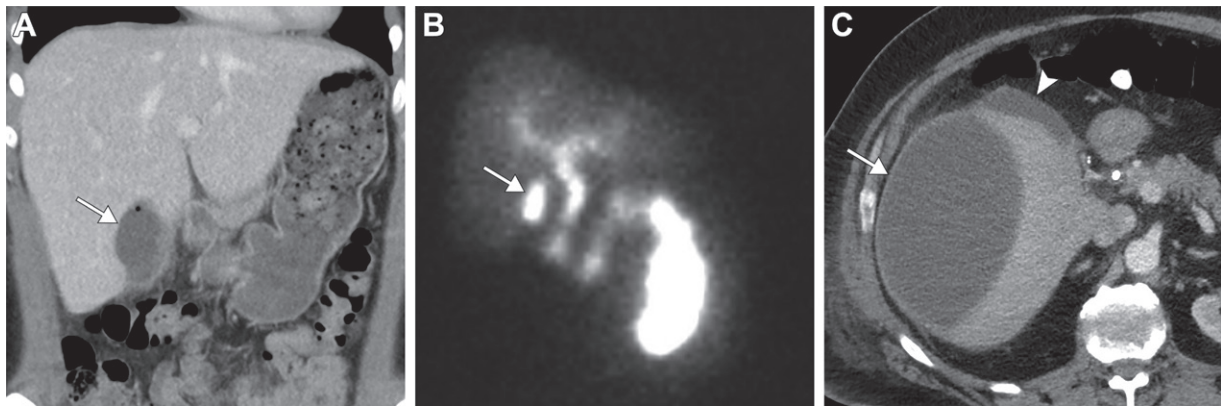


Figure 6. Biloma in two patients. (A) Coronal contrast-enhanced CT image in a 28-year-old woman with right upper quadrant pain 14 days after laparoscopic cholecystectomy shows an oval fluid collection (arrow) in the gallbladder fossa, with an irregular enhancing wall and a small amount of gas. (B) Left anterior-oblique ^{99m}Tc HIDA scintigraphic image in the same patient as in A at 30 minutes after administration of contrast material shows the accumulation of tracer in the gallbladder fossa collection (arrow), which is consistent with a biloma. (C) Axial contrast-enhanced CT image in a 65-year-old man with pain and shortness of breath 13 days after laparoscopic cholecystectomy shows a large hepatic subcapsular fluid collection (arrow) that is connected to a smaller gallbladder fossa collection (arrowhead), which is most typical of and was confirmed to be a biloma after percutaneous drainage.

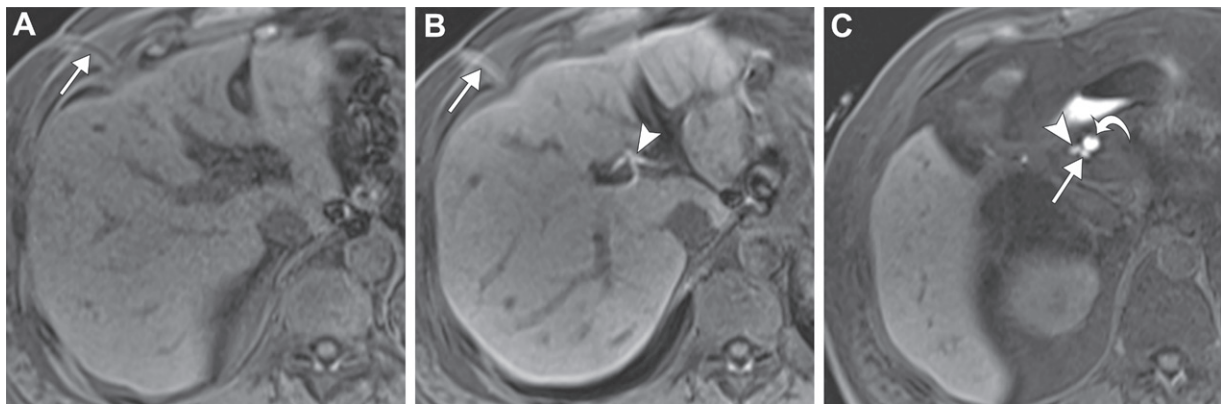


Figure 7. Bile leak in a 57-year-old woman with worsening drain output after complicated cholecystectomy. (A) Axial precontrast T1-weighted fat-saturated gradient-recalled-echo MR image shows a hypointense lumen of the percutaneous biliary drain (arrow). (B) Axial gadoxetate-enhanced T1-weighted fat-saturated MR image at 20 minutes after administration of contrast material shows opacification of the central intrahepatic ducts (arrowhead) and the drain lumen (arrow), which are findings consistent with a bile leak. (C) Axial T1-weighted fat-saturated MR image acquired 20 minutes after administration of contrast material shows the site of the leak (straight arrow), emanating from the junction of the cystic duct (arrowhead) and the common duct (curved arrow).

delay. Therefore, if there is high clinical suspicion for a bile leak, a delay of 60–90 minutes after contrast administration should be considered if the 20-minute image is inconclusive. The hepatobiliary phase contrast-enhanced images are particularly helpful in patients with a biloma, which may coexist with other injuries and, if large enough, can make MRCP images difficult to interpret because of overlapping and obscuring anatomy (29).

Heavily T2-weighted MRCP sequences must be performed before administering gadoxetate because it is excreted into the bile ducts as early as 5–10 minutes after administration and results in T2 shortening and loss of the signal intensity of fluid in the bile ducts, reducing the diagnostic quality of the MRCP images (26,30).

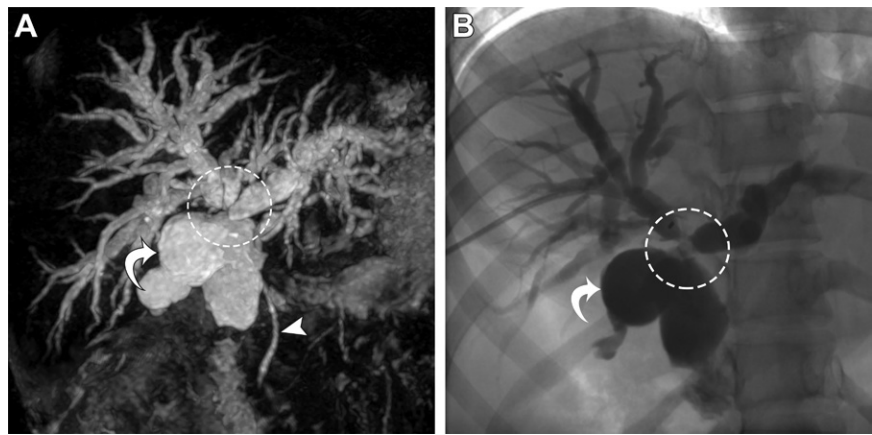
Bile leaks and bilomas are treated first with drainage of fluid to prevent secondary infec-

tion and to establish adequate biliary drainage. Percutaneous transhepatic cholangiography or endoscopic retrograde cholangiopancreatography (ERCP) is then usually performed to define the anatomy of the leak (Fig 5). Leaks may also be treated during ERCP with biliary stenting, with or without sphincterotomy, to reduce the pressure gradient across the leak. More extensive injuries may require surgical intervention, such as Roux-en-Y hepaticojejunostomy (15,23,31).

Acute Biliary Obstruction

Biliary injuries near the common bile duct or the confluence of hepatic ducts may acutely obstruct the biliary tree. Patients may present with jaundice, pruritis, or ascending cholangitis (9). These injuries require a high index of suspicion because they mimic a retained stone (ie, choledocholithia-

Figure 8. Hilar bile duct injury in a 21-year-old woman with worsening jaundice after laparoscopic cholecystectomy. (A) Coronal-oblique three-dimensional maximum intensity projection MR cholangiopancreatographic (MRCP) image shows dilated intrahepatic bile ducts converging on three separate hilar strictures of the right anterior, right posterior, and left hepatic ducts (dashed circle). A large biloma is also seen obscuring some of the anatomy (curved arrow). A normal common bile duct can be seen distally (arrowhead). (B) Left posterior-oblique fluoroscopic image from percutaneous cholangiography after access of the right biliary system shows the hilar biliary strictures (dashed circle) with adjacent surgical clips and filling of the biloma (curved arrow). Notice that only two of the three biliary strictures are identified, and some dilated ducts seen on the MRCP image in A are nonopacified, which highlights the importance of multimodality imaging. The patient underwent hepaticojejunostomy, with three separate anastomoses.



sis) at clinical presentation and may be missed at the time of surgery. CT and US show either segmental or diffuse intrahepatic biliary duct dilatation. CT readily allows identification of the location of metallic surgical clips, which may have been placed across the injury, although it is often difficult to identify which clip is the offending one. Bilomas often coexist and may confound interpretation, particularly on MRCP images. A vascular injury may coexist with a biliary injury, such as occlusion of the right hepatic artery, and multiphasic CT of the liver and conventional angiography are often useful for detection.

MRI can add better anatomic detail to the biliary system and can help identify the level of biliary obstruction (Fig 8). Hepatobiliary scintigraphy will show delayed biliary excretion to the level of the obstruction and no tracer in the duodenum. If there is increased pressure in the biliary duct, there may be little to no biliary excretion (32,33). ERCP can be performed but is often of limited diagnostic utility because the obstruction can be difficult to cross and the intrahepatic bile ducts may not be opacified with contrast material (31). Percutaneous cholangiography is often the initial procedure of choice because it allows identification of the biliary anatomy, localization of the injured segment or segments, and treatment with internal and/or external biliary drainage. A multimodality approach including CT, MRI, and cholangiography is often necessary to help the surgeon plan for a biliary reconstruction, because many patients require a biliary diversion procedure (15,31).

Biliary Stricture

Biliary strictures are a rare delayed complication of cholecystectomy that manifest months or years after surgery (9,34,35). Patients can pres-

ent with nonspecific pain, jaundice, pruritis, or infection. Unrecognized and untreated central biliary strictures can result in secondary biliary cirrhosis (34). Similarly, segmental strictures may result in atrophy of a portion of the liver. Strictures develop because of fibrosis adjacent to surgical clips, after direct injury, or after thermal injury during dissection. Like acute injuries, variant biliary anatomy is a risk factor for biliary stricture because of inadvertent injury or ligation of an anomalous duct.

One of the most commonly encountered biliary strictures after cholecystectomy is aberrant stricturing of the right posterior hepatic duct. This duct is prone to injury because it often inserts on either the cystic duct or the common duct near the cystic duct, leading to direct clipping or injury. This injury has a specific pattern on cross-sectional images, with segmental dilatation of posterior right intrahepatic ducts, and the dilated aberrant duct often can be traced to its low insertion on the common duct (Fig 9). If the stricture is chronic and of high grade, segmental hepatic atrophy can be encountered as volume loss of segments VI and VII (Fig 10). Some of these injuries can be conservatively managed, whereas others may require liver resection or biliary drainage (16,36).

Stone-related Complications

Retained and Recurrent Stones

Retained stones are stones that are unrecognized during surgery, whereas recurrent stones develop in the extrahepatic biliary system after surgery. Intraoperative cholangiography is highly sensitive for detecting retained stones during surgery, but it is not performed routinely because it has not been shown to prevent biliary injuries (37). If a

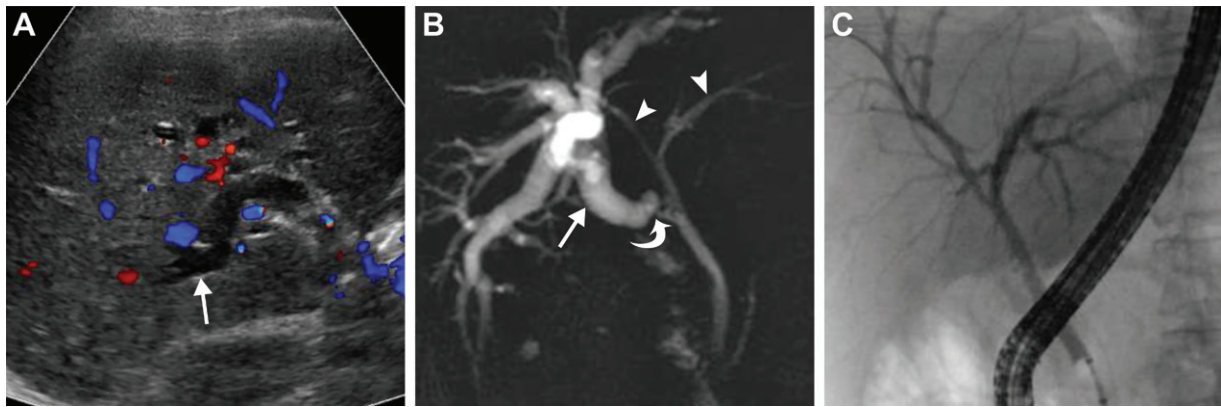


Figure 9. Aberrant stricture of the right hepatic duct in a 55-year-old woman with a history of remote cholecystectomy 20 years earlier who presented with abdominal pain and elevated liver function test results. **(A)** Transverse US image of the liver shows dilated bile ducts (arrow) in the right lobe. **(B)** Two-dimensional MRCP image shows severe isolated biliary dilatation of the right posterior ducts (straight arrow), with normal right anterior and left hepatic ducts (arrowheads). The right posterior ducts insert aberrantly on the cystic duct, where there is a stricture (curved arrow). **(C)** Anterior fluoroscopic image from ERCP shows a normal right anterior and left biliary tree with nonfilling of the dilated right posterior ducts, which confirms occlusion secondary to surgical injury during the previous cholecystectomy.

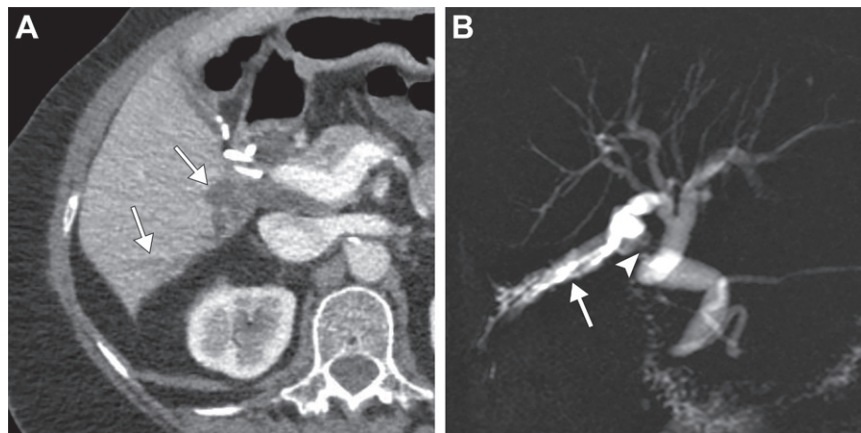


Figure 10. Aberrant stricture in the right hepatic duct with liver parenchymal atrophy in a 69-year-old woman with abnormal liver function test results and a history of remote cholecystectomy 20 years earlier. **(A)** Axial contrast-enhanced CT image shows isolated intrahepatic biliary dilatation in the posterior right lobe (arrows) that is associated with parenchymal atrophy of segments VI and VII. **(B)** Coronal-oblique two-dimensional MRCP image shows clustered dilated right posterior segment bile ducts (arrow) with stricturing (arrowhead) at the low insertion on the common duct or cystic duct as a result of biliary injury at the time of the cholecystectomy. The dilatation of the common, right anterior, and left biliary ducts was thought to be unrelated.

stone is discovered less than 3 years after surgery, some authors (38) consider it more likely to be a retained stone. However, distinguishing between these two entities is not always possible or clinically necessary, and temporal definitions vary in the literature (38,39).

Choledocholithiasis is found in 3.4% of patients who undergo a laparoscopic cholecystectomy, according to a prospective study of 997 patients (40). The incidence of retained stones in the common bile duct after cholecystectomy is 0.5% (8). The incidence of retained stones in other biliary locations is not well defined. The clinical presentation is vague and can range from mild abdominal pain to symptoms of biliary ob-

struction (41). While most patients present soon after surgery, some may not present until months or years after the initial surgery. The stone can cause distention and inflammation of the remnant cystic duct (42). If the stone grows in the cystic duct, it may cause obstruction or mass effect on the common bile duct, known as Mirizzi syndrome (Fig 11) (43). The stone in the cystic duct can also dislodge into and obstruct the common bile duct. Rarely, a chronically retained stone can migrate through a biliary-duodenal fistula to the duodenum and result in bowel obstruction, as in gallstone ileus or Bouveret syndrome. The risk factors for retained or recurrent stones include subtotal cholecystectomy, biliary

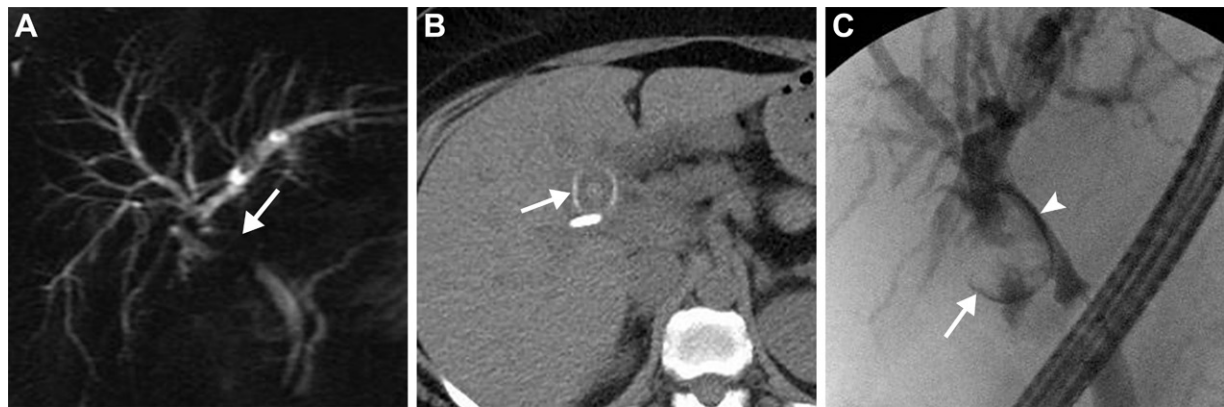


Figure 11. Retained stone in the cystic duct in a 43-year-old woman with jaundice and Mirizzi syndrome on day 6 after laparoscopic cholecystectomy. (A) Two-dimensional MRCP image shows narrowing in the long segment of the upper common duct (arrow), with mild upstream biliary dilatation. (B) Axial noncontrast CT image shows a large lamellated stone (arrow) in the gallbladder fossa adjacent to a surgical clip and external to the common duct. Mirizzi syndrome was suspected. (C) Anterior fluoroscopic image from ERCP shows a large filling defect in the cystic duct (arrow) that is consistent with a retained stone. Note the smooth external compression on the common duct (arrowhead) and upstream biliary dilation, findings that help confirm Mirizzi syndrome.

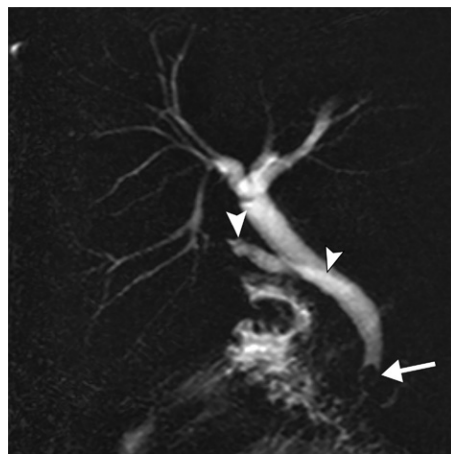


Figure 12. Recurrent stone disease in a 40-year-old man with elevated liver function test results and abdominal pain 20 years after undergoing cholecystectomy. Coronal-oblique two-dimensional MRCP image shows mild biliary dilatation due to distal choledocholithiasis (arrow). Notice the low insertion of a long cystic duct stump (arrowheads) which may have contributed to stone formation by means of stasis.

stasis, multiple common bile duct stones at initial presentation, or clip migration, which acts as a nidus for stone formation (16,36,41). The most common locations for retained stones, in decreasing order of frequency, are the remnant gallbladder, the cystic duct, and the common bile duct. Treatment involves removing the stone, typically endoscopically (41).

Imaging appearances of a retained or recurrent stone in the common bile duct are similar to those of typical cholelithiasis or choledocholithiasis, with a filling defect in the cystic or common ducts (Fig 12). Retained or recurrent stones can cause secondary complications including biliary obstruction or cholangitis (41).

Spillage of Gallstones

Gallbladder perforation and subsequent gallstone spillage into the peritoneum (ie, “dropped stones”) occurs in up to 40% of laparoscopic cholecystectomies in some studies (44,45). If spillage occurs during surgery, the surgeon should attempt to retrieve the stones and irrigate the abdomen (45). The incidence of unretrieved stones is uncertain. The perihepatic space, specifically the hepatorenal fossa, is the most common location of a spilled (or “dropped”) stone, although the stones can be found throughout the abdomen and pelvis and have been reported in the thorax (44,45).

At imaging, the appearance of dropped gallstones can vary with stone composition, stone size, and number of stones spilled. Calcified stones can be seen on radiographs and are hyperattenuating on CT images. Cholesterol stones or pigmented stones often appear as non-specific small, round, soft-tissue nodules with intermediate attenuation at CT, and if they are numerous, they can mimic peritoneal carcinomatosis because they are typically clustered in the right upper quadrant. Dropped gallstones can elicit a local granulomatous reaction, which manifests on fluorine 18 (^{18}F) fluorodeoxyglucose (FDG) PET images as hypermetabolic peritoneal nodules, also mimicking peritoneal carcinomatosis (46). Dropped stones can mimic masses in other organs such the stomach or urinary bladder dome (Fig 13). When they are implanted on the liver, dropped stones may mimic metastatic disease or other liver lesions (Fig 14). MRI can be helpful in these cases, with stones showing hypointensity with T1- and T2-weighted sequences and nonenhancement, although pigment stones can be T1 hyperintense

Figure 13. Dropped gallstone mimicking a gastric mass in a 44-year-old woman with a history of cholecystectomy in the remote past who presented with vague abdominal pain. (A) Endoscopic image of the distal stomach shows a bulging lesion with normal overlying mucosa (arrow). (B) Axial contrast-enhanced CT image obtained afterward to look for an extrinsic mass shows a large lamellated dropped gallstone (arrow) in the same location as the endoscopic abnormality.

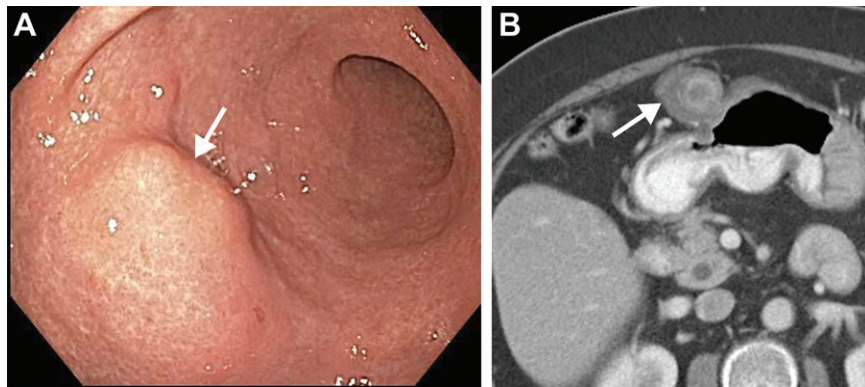
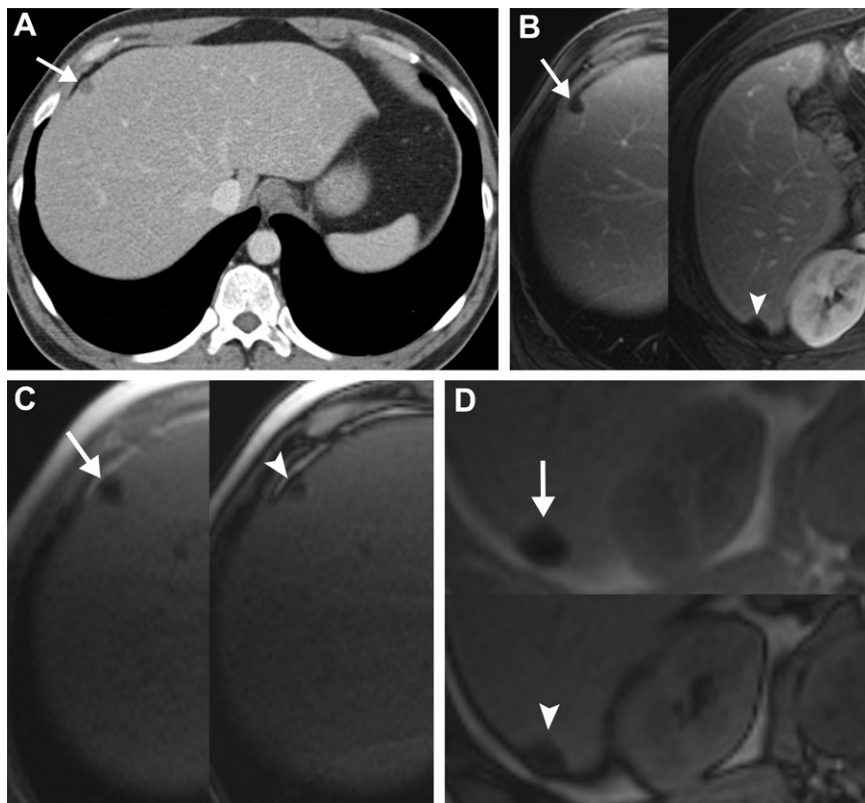


Figure 14. Dropped gallstones mimicking liver metastases in a 43-year-old man with testicular cancer. (A) Axial contrast-enhanced CT image from the initial staging examination shows a small hypoattenuating mass (arrow) near the surface of segment VIII. A similar lesion was seen in segment VI, and therefore, MRI was recommended. (B) Axial T1-weighted fat-saturated contrast-enhanced MR images show no enhancement in the segment VIII lesion (arrow) or the segment VI lesion (arrowhead). (C) In-phase and out-of-phase T1-weighted MR images of the segment VIII lesion show blooming of the lesion on the in-phase image (arrow) compared with the out-of-phase image (arrowhead). (D) In-phase and out-of-phase T1-weighted images of the segment VI lesion show the same blooming on the in-phase image (arrow) compared with the out-of-phase image (arrowhead). The patient had undergone prior cholecystectomy, and therefore, dropped gallstones were diagnosed.



(47). T1-weighted in- and out-of-phase gradient-echo MR images are especially helpful for identifying gallstones because stones often show blooming on in-phase images because of susceptibility artifact (48).

Despite their high prevalence, most dropped gallstones are asymptomatic. Although the reported range of complications from dropped gallstones in the literature is highly variable (0.04%–19%), the largest studies to date show rates of less than 0.3% (44). Complications may occur soon after surgery or may manifest months or years after surgery. The most common complication of a dropped stone is formation of an abscess, which classically is found in the hepatorenal space (right subhepatic space) (Fig 15). Other patients present with a phlegmon, and sinus tracts to the skin or bowel can form in

rare cases because the infection may cross fascial planes. Some infections may follow a more chronic or indolent clinical course, and patients may not present with symptoms of illness (44,47).

Although it is not always possible to identify the stone within the abscess or phlegmon, careful evaluation for a stone nidus should be performed (Figs 16, E1). US can be particularly helpful because it may show a hyperechoic nodule with posterior acoustic shadowing similar to gallstones in the gallbladder (Fig 15). Even in the absence of an identified stone nidus at imaging, an unexplained abscess or inflammatory mass in the hepatorenal fossa in a patient with a history of cholecystectomy should be considered highly suspicious for a dropped gallstone as the underlying cause. The radiologist may be the first to

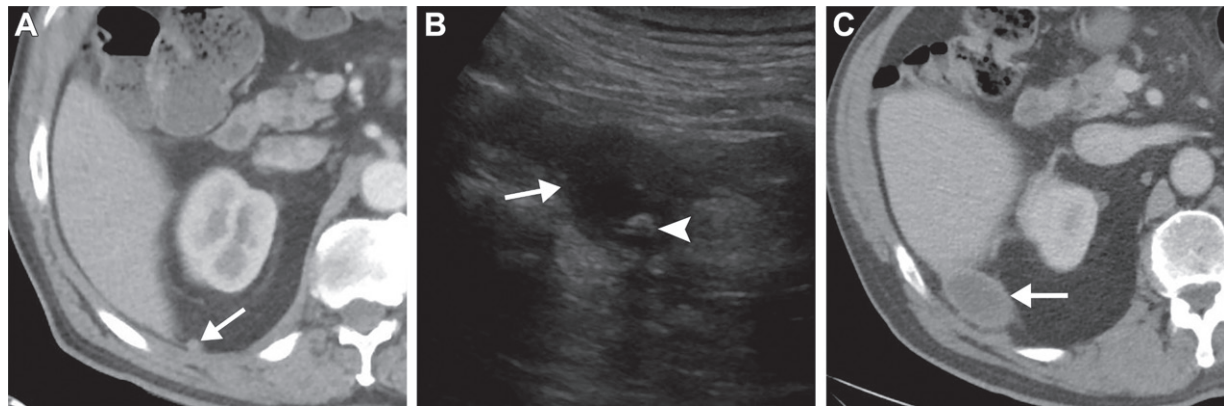


Figure 15. Dropped gallstone with an abscess in a 68-year-old man 3 years after a laparoscopic cholecystectomy. (A) Axial contrast-enhanced CT image obtained 1 year after surgery shows a small round lesion in the right upper quadrant adjacent to the liver edge (arrow) that is typical of a dropped gallstone. (B) Transverse US image acquired 3 years after surgery in the area of pain in the right upper quadrant shows a round fluid collection (arrow) containing an echogenic focus (arrowhead) with posterior acoustic shadowing. (C) Subsequent axial contrast-enhanced CT image shows a rim-enhancing fluid collection (arrow) at the same site, a finding that is consistent with an abscess complicating the dropped gallstone. The patient underwent surgical drainage and stone extraction.

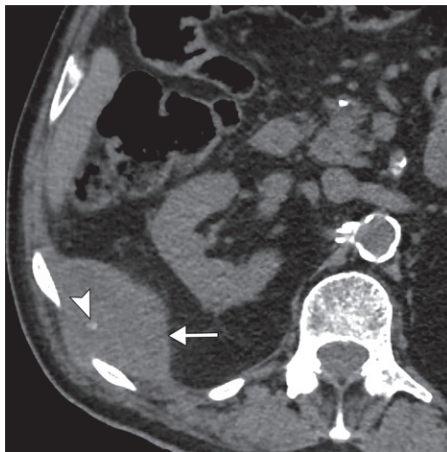


Figure 16. Dropped gallstone with abscess in a 90-year-old woman with a remote history of laparoscopic cholecystectomy. Axial noncontrast CT image shows a fluid collection in the right paracolic gutter (arrow) crossing fascial planes into the right retroperitoneum and through the chest wall surrounding the ribs. The small round hyperattenuating focus in the center of the collection (arrowhead) is typical of a dropped gallstone acting as the nidus of infection and was the clue to accurate diagnosis.

suggest this diagnosis, which alters the treatment strategy because it generally involves removing the underlying stone, which, if left in situ, could serve as a source of recurrent infection (47).

Iatrogenic Injuries

Hemorrhage

Postoperative hemorrhage is reported to occur in 0.79% of laparoscopic cholecystectomies and is usually a result of mechanical or thermal injury during surgery (8,16). Bleeding complications may not be recognized intraoperatively because of

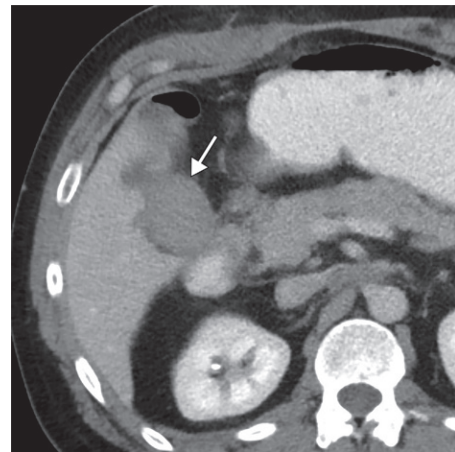
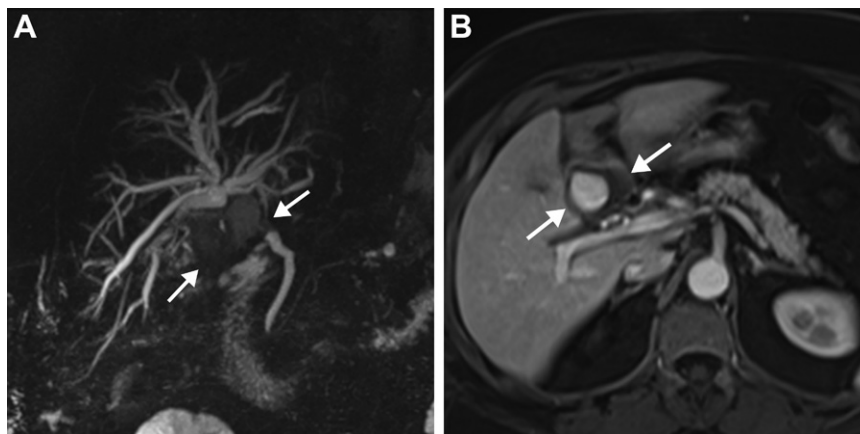


Figure 17. Postoperative hematoma in a 51-year-old man with acute pancreatitis and choledocholithiasis on day 10 after laparoscopic-converted-to-open cholecystectomy. Axial contrast-enhanced CT image shows a hyperattenuating collection in the gallbladder fossa (arrow) with an average attenuation value of 65 HU, which is typical of a hematoma.

patient hypotension during surgery or tamponade from the pneumoperitoneum created for laparoscopic surgery (49). The most commonly injured vessels are the right hepatic artery and the portal vein, but bleeding can arise from the liver parenchyma in the gallbladder bed or the cystic artery (16). Most commonly, acute hemorrhage collects in the gallbladder fossa, which may be mistaken for a biloma or abscess. CT is the best modality to evaluate for postoperative hematoma because attenuation values of 40–60 HU can usually help distinguish a hematoma from other postoperative fluid collections (Fig 17), for which attenuation values are typically lower. However, the appearance may vary, depending on the age of blood products and the admixture of blood with bile or

Figure 18. Cystic artery pseudoaneurysm in a 54-year-old woman with elevated values on liver function test results several months after laparoscopic cholecystectomy. **(A)** MRCP image shows mild intrahepatic biliary dilatation and absence of signal intensity in the upper common duct, with a rounded void of signal intensity (arrows) at the level of the stricture. **(B)** Axial contrast-enhanced fat-saturated T1-weighted MR image shows that the MRCP signal intensity void corresponds to a large pseudoaneurysm (arrows) of the cystic artery, with a mural thrombus, which caused mass effect on the bile ducts.



pus. In some cases, depending on the imaging appearance, the differential diagnosis at CT should include evaluation for biloma or abscess. Further imaging with HIDA scintigraphy or hepatobiliary phase MRI may add specificity to the imaging diagnosis through evaluation for a bile leak.

Vasculobiliary Injury

Vasculobiliary injury represents a concomitant injury to a bile duct and a hepatic artery and/or a portal vein (50). The most common injury is to the right hepatic artery (Fig E2) because of its anatomic proximity to the common hepatic duct, which is mistakenly divided during cholecystectomy. The true incidence of this injury is unknown because routine angiography is not performed. One review (50) reported that 12%–61% of patients with a bile duct injury had a concurrent right hepatic artery injury. Isolated right hepatic artery occlusion during cholecystectomy is thought to occur frequently, but because of collateral flow along the bile duct, patients are usually asymptomatic, without clinically significant biliary ischemia (50). However, concurrent biliary injury affects this collateral arterial flow, most severely with hilar injuries, which may result in bile duct necrosis occurring over several months. In approximately 10% of cases, slow infarction of the right hepatic lobe occurs over several weeks, resulting in areas of necrosis or eventual atrophy. In rare cases, injury to the portal vein, common hepatic artery, or proper hepatic artery occurs, which may result in more rapid infarction (Fig E3), requiring urgent right hepatectomy or a liver transplant (50). Because of this high association of injuries, multiphasic liver imaging is often used to evaluate for vascular injury whenever a biliary injury is suspected.

Arterial Pseudoaneurysm

Cystic or hepatic artery pseudoaneurysm is a rare complication after cholecystectomy and may manifest in the early postoperative period or be delayed

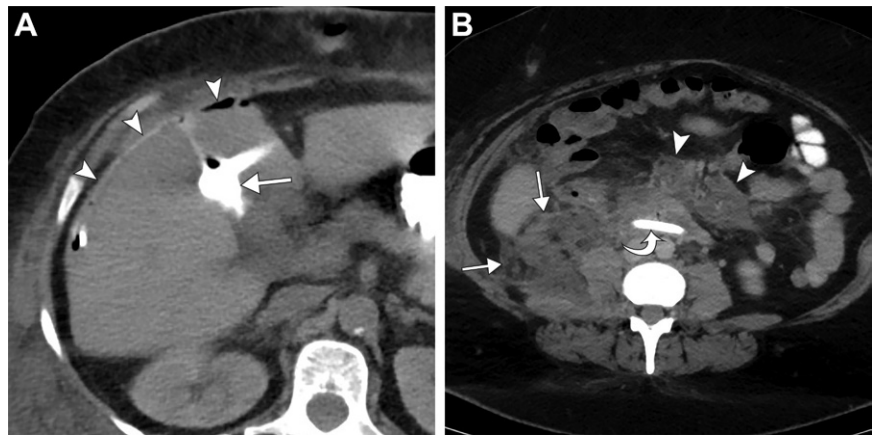
for multiple months. Patients may present with the Quincke triad: jaundice, right upper quadrant pain, and upper gastrointestinal bleeding, although the clinical presentation can vary (51–53). The underlying mechanism for formation of a pseudoaneurysm is uncertain, but it is hypothesized that local inflammation and fibrosis combined with electrocauterization during surgery injures the arterial wall, leading to a pseudoaneurysm (51,52). Bile leak is a risk factor for formation of a pseudoaneurysm, because bile can damage the vascular wall and predispose patients to formation of a pseudoaneurysm (54).

At CT angiography, a pseudoaneurysm shows a rounded, often heterogeneous mass in the porta hepatis that enhances avidly during the arterial phase and follows the blood pool in later phases of imaging. MRCP may be performed because of jaundice and suspicion for a retained stone. The pseudoaneurysm may be seen as a signal void in the hilum. T2-weighted MR images have a variable appearance, depending on the size of the pseudoaneurysm and whether flow voids or a mural thrombus is present. Contrast-enhanced T1-weighted MR images show a variable degree of central enhancement, which is most intense during the arterial phase. The wall is often of variable thickness and may enhance during the delayed phase. A mural thrombus may be seen, depending on the age of the pseudoaneurysm (Fig 18). Treatment usually involves transarterial embolization or surgical ligation (51,52).

Duodenal Injury

Duodenal injuries directly from cholecystectomy are rare but are associated with high morbidity and mortality rates. Injury may occur during cautery, dissection, or retraction or from the use of laparoscopic trocars (55,56). As with other complications, the symptoms are vague and nonspecific, including increasing postoperative abdominal pain, fever, nausea, and vomiting. Duodenal injury may also occur during ERCP, before or

Figure 19. Duodenal injury in two patients. **(A)** Intraperitoneal duodenal injury in a 67-year-old woman 4 days after she underwent a complicated cholecystectomy for perforated cholecystitis. Axial noncontrast CT image shows oral contrast material and gas leaking into the gallbladder fossa (arrow) and tracking around the edge of the liver (arrowheads). The patient subsequently underwent surgery for repair of iatrogenic duodenal bulb injury. **(B)** Retroperitoneal duodenal injury in a 32-year-old woman after a complicated laparoscopic-converted-to-open cholecystectomy. Axial contrast-enhanced CT image shows large heterogeneous fluid collections in the right pararenal space (straight arrows) and central mesentery (arrowheads) surrounding the duodenum and containing a feeding tube (curved arrow). The collections appear similar to those encountered in patients with necrotizing pancreatitis.



after cholecystectomy, and is more common than a surgical injury (57). Early detection of duodenal injuries is important because this portends better survival (55). Contrast-enhanced CT is the best modality for identification of duodenal injury. Findings include periduodenal or retroperitoneal fluid, intraperitoneal or retroperitoneal gas, oral contrast material leakage, and extensive fat stranding (Fig 19). It is important to remember that duodenal perforations may be retroperitoneal; and classically, extraluminal gas and fluid extend inferiorly in the right anterior pararenal space. These imaging findings often overlap with those of acute pancreatitis, which may coexist with cholecystitis or be a complication of ERCP; however, in general, a duodenal injury spares the body and tail of the pancreas, and fluid collections are confined to the right retroperitoneum. The addition of oral contrast material is an important tool when evaluating for duodenal injury, because extraluminal oral contrast material is a specific finding that can help distinguish duodenal injury from other causes of periduodenal fluid and inflammation. Upper gastrointestinal fluoroscopic imaging can also be used to detect a duodenal injury, and the dynamic nature of the examination can allow evaluation of the volume of the leak and whether it is locally contained or disseminated (Fig E4).

Clip Migration

Clip migration is a rare complication with fewer than 100 cases reported between 1978 and 2008 (58). The underlying mechanism for clip migration is poorly understood. One proposed mechanism is that cystic duct necrosis results in erosion of the clip into the lower-pressure biliary tree (59). Risk factors for clip migration include the use of four or more clips, local inflammation in the clipped area, or a subtotal cholecystectomy (60). Patient

presentation with clip migration is usually delayed, with a median time to presentation of 2 years (58). The most common location to detect a migrated clip is in the common bile duct, where it can cause biliary obstruction or act as a nidus for formation of choledocholithiasis (38,59).

At imaging, radiographs show a clip displaced inferiorly, away from the other clips in the liver hilum. CT shows a metallic filling defect in the common bile duct. MRI findings are generally indistinguishable from those of a stone, including a T2-hypointense filling defect and associated susceptibility artifact (Fig 20). Comparing the number and location of the clips to those in prior examinations is often helpful. Treatment usually involves ERCP with cannulation and sweeping of the common duct, which is similar to the treatment for choledocholithiasis.

Gallbladder Complications

Recurrent Cholecystitis after Subtotal Reconstituting Cholecystectomy

Recurrent symptomatic cholelithiasis or cholecystitis is one of the risks of a subtotal reconstituting cholecystectomy, although the incidence and long-term outcomes of subtotal cholecystectomies are not clearly defined (10,61). The clinical symptoms are similar to those of conventional symptomatic cholelithiasis or cholecystitis (62,63). Furthermore, the radiologist may be the first practitioner to suspect this diagnosis because many patients who have undergone subtotal cholecystectomy may be erroneously labeled in the medical record or may self report having undergone a traditional cholecystectomy; therefore, a diagnosis of cholecystitis would not be suspected. The amount of native gallbladder removed during a subtotal reconstituting cholecystectomy is variable (10).

Figure 20. Cholecystectomy clip migration in a 50-year-old woman with a history of remote cholecystectomy 17 years earlier who presented with pain and elevated liver function test results. **(A)** Coronal three-dimensional MRCP image shows a filling defect in the distal common bile duct with the meniscus sign (arrow), resulting in dilatation of the biliary duct and cystic duct stump (arrowhead). **(B)** Coronal contrast-enhanced maximum intensity projection CT image shows that the filling defect (arrow) is a metallic foreign body that is similar in size and shape to the two other surgical clips in the hilum (arrowheads). **(C)** Coronal contrast-enhanced CT image from 2 years earlier shows that there were previously three clips (arrowheads) in the liver hilum. Therefore, clip migration was suspected. **(D)** Posteroanterior fluoroscopic image obtained during ERCP to confirm removal of the foreign body after sweeping a stone into the duodenum shows the metallic clip (arrow) encrusted by stone (arrowhead), lying in the duodenum, with an overlying hemostat on the skin for reference.

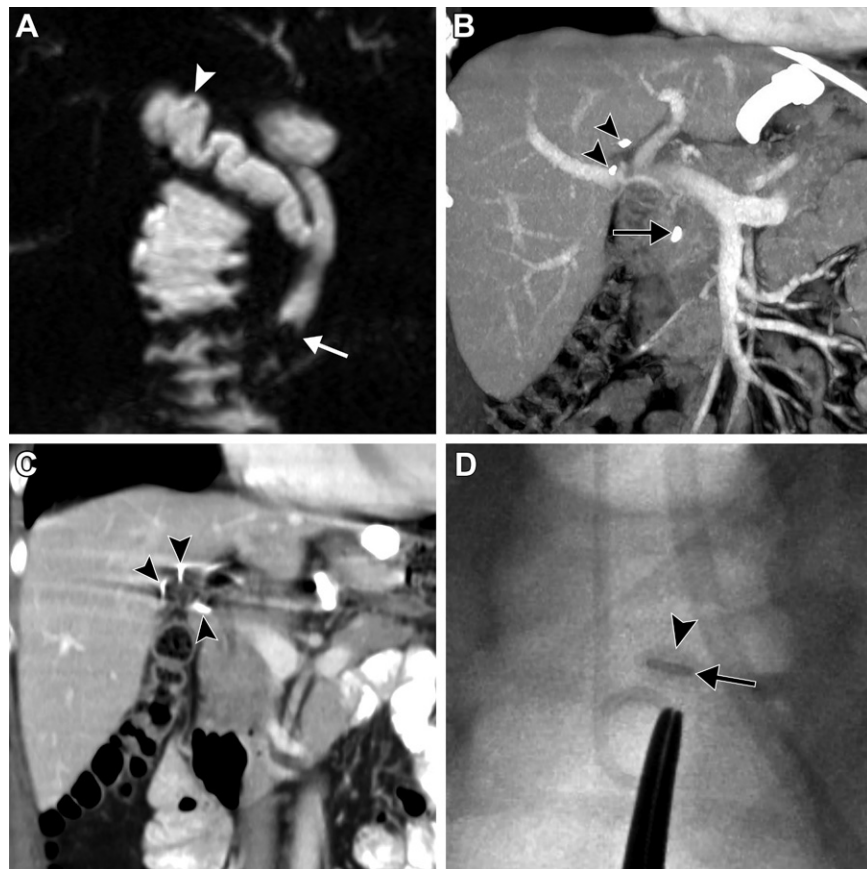
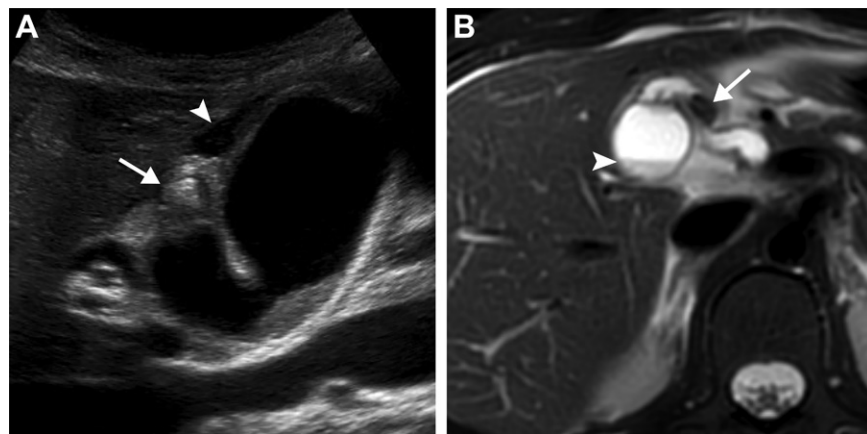


Figure 21. Recurrent cholecystitis after subtotal cholecystectomy in a 25-year-old man with Crohn disease and a history of subtotal cholecystectomy 3 years earlier who presented with right upper quadrant pain. **(A)** Longitudinal US image shows a gallbladder containing sludge and a stone that is lodged in the neck (arrow). The mild distention, pericholecystic fluid (arrowhead), and gallbladder wall thickening were consistent with recurrent acute cholecystitis, despite the history of cholecystectomy. **(B)** Axial T2-weighted fat-saturated MR image shows the stone in the gallbladder neck (arrow), layering sludge (arrowhead), and pericholecystic edema. The patient underwent cholecystectomy, and surgical pathologic results confirmed the diagnosis of acute cholecystitis.



US, CT, and MRI findings are similar to those in patients with acute cholecystitis, although the gallbladder may be smaller (Fig 21). Treatment of recurrent cholecystitis is ultimately a completion cholecystectomy (11,62).

Unsuspected Gallbladder Cancer

In most surgical literature, an unexpected diagnosis of gallbladder cancer after cholecystectomy for benign disease is not typically classified as a surgical complication. However, we believe it

is worth including because the radiologist may be the first to suspect the underlying diagnosis and prevent the wrong surgery from being performed, because surgery for suspected cancer often involves a more radical resection with liver resection. Unsuspected gallbladder cancer occurs in 0.7% of patients who undergo cholecystectomy in one meta-analysis (64). In comparison, the overall incidence of gallbladder cancer in the United States is 1–1.5 per 100 000 individuals (65,66). Preoperative diagnosis is

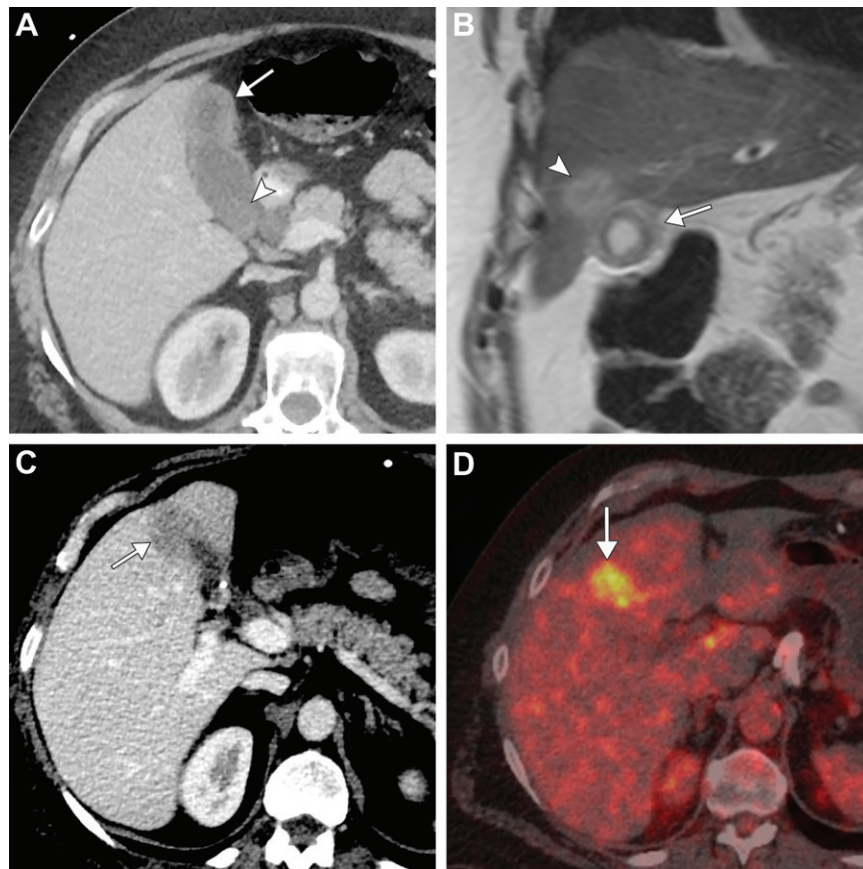


Figure 22. Unsuspected gallbladder cancer in a 57-year-old man with acute right upper quadrant pain. (A) Contrast-enhanced CT image shows focal severe wall thickening of the fundus of the gallbladder (arrow), sparing the body and neck. Note the lack of adjacent fat stranding. A subtle gallstone is also seen at the neck (arrowhead). (B) Coronal single-shot fast spin-echo MR image obtained the same day shows the focal irregular thickening of the gallbladder wall (arrow) and an adjacent round hyperintense lesion in segment IVb of the liver (arrowhead). The diagnosis was incorrectly determined to be acute cholecystitis with a liver abscess. A simple laparoscopic cholecystectomy was performed, and pathologic results showed adenocarcinoma of the gallbladder with positive margins. (C) Postoperative contrast-enhanced CT image shows a persistent hypoenhancing mass in the liver (arrow) adjacent to the gallbladder fossa. (D) ^{18}F FDG PET/CT image shows focal uptake in the lesion (arrow). The patient subsequently underwent liver resection.

difficult because there are many similarities in risk factors, clinical presentation, and imaging appearance between gallbladder cancer and other more common indications for cholecystectomy. Gallbladder cancer is usually asymptomatic until advanced, but symptoms can mimic biliary colic (66). Preoperative imaging diagnosis is difficult because many of the imaging patterns mimic acute or chronic cholecystitis or adenomyomatosis (66). At imaging, gallbladder cancer manifests as a mass arising from the wall or replacing the gallbladder lumen in 40%–70% of cases, focal or diffuse asymmetric wall thickening in 20%–30% of cases, or a polypoid lesion in 15%–25% of cases. On CT images, a heterogeneously enhancing and thickened gallbladder wall or a stratified enhancement pattern with a hyperenhancing inner mucosal layer are statistically more likely to represent gallbladder cancer than is benign wall thickening (67). Reported dynamic contrast-enhanced MRI findings favoring malignancy over benign gallbladder abnormalities include irregular or focal early enhancement, which persists with delayed imaging (68). Radiologists should consider possible gallbladder cancer if there is focal, irregular, or asymmetric wall thickening and the absence of adjacent fat stranding, and they also should consider local tumor extension with an adjacent liver lesion (Fig 22) (66).

Conclusion

Cholecystectomy is associated with several early and delayed complications. Many of these complications have vague and nonspecific clinical signs and symptoms and can result in morbidity and mortality if unrecognized. Consequently, multimodality imaging is important in the evaluation of these complications. Familiarity with the imaging manifestations can assist the referring providers with accurate diagnosis and treatment.

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